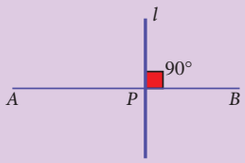
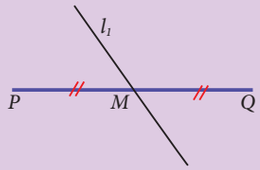
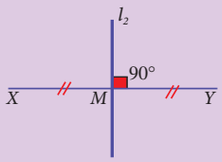
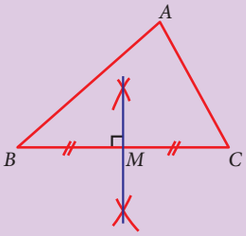
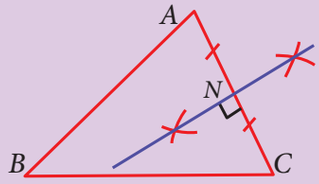
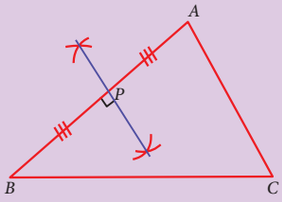
5.8 Perpendicular Bisectors of a Triangle

Let us first recall the following ideas.

Perpendicular	Bisector	Perpendicular bisector
		
$\overline{AB}$ is a line segment. l is perpendicular to $\overline{AB}$. P is the foot of the $\perp^r$. Note that $\overline{AP} \uparrow \overline{PB}$ here.	$\overline{PQ}$ is a line segment. l_1 is a bisector to $\overline{PQ}$. M is the midpoint of $\overline{PQ}$. l_1 need not be $\perp^r$ to $\overline{PQ}$	$\overline{XY}$ is a line segment. l_2 is a bisector to $\overline{XY}$. l_2 is also $\perp^r$ to $\overline{XY}$. M is the midpoint of $\overline{XY}$.

Consider a triangle ABC. It has three sides. For each side you can have a perpendicular bisector as follows:

		
Perpendicular bisector of side BC	Perpendicular bisector of side AC	Perpendicular bisector of side AB
M is the midpoint of $\overline{BC}$ We have a perpendicular at M.	N is the midpoint of $\overline{AC}$ We have a perpendicular at N.	P is the midpoint of $\overline{AB}$ We have a perpendicular at P.

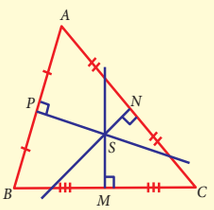
Surprisingly, all the three perpendicular bisectors of the sides of a triangle are concurrent at a point!

Activity

One can visualize the point of concurrence of the perpendicular bisectors, through simple paper folding. Try and see that,

The perpendicular bisectors of the sides of *any* triangle are concurrent.

As done in the earlier activity on Centroid, you can repeat the experiment for various types of triangle, acute, obtuse, right, isosceles and equilateral. Do you find anything special with the equilateral triangle in this case?



5.8.1 Circumcentre

The point of concurrence of the three perpendicular bisectors of a triangle is called as its **Circumcentre,** denoted by the letter **S**.

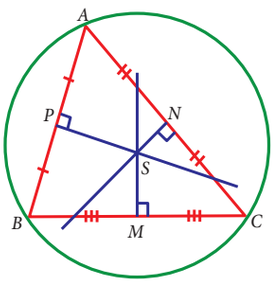


Fig. 5.35

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